

MODULE 3

Aggregate Planning

Roles & Need for Aggregate Planning

Methods of Aggregate Planning

Graphical Method
Reaction Method

Aggregate Planning

It's a marketing activity that does an aggregate plan for the production process, in advance of 6 to 18 months, to give an idea to management as to

- what quantity of materials and other resources are to be procured
- When to be procured ;

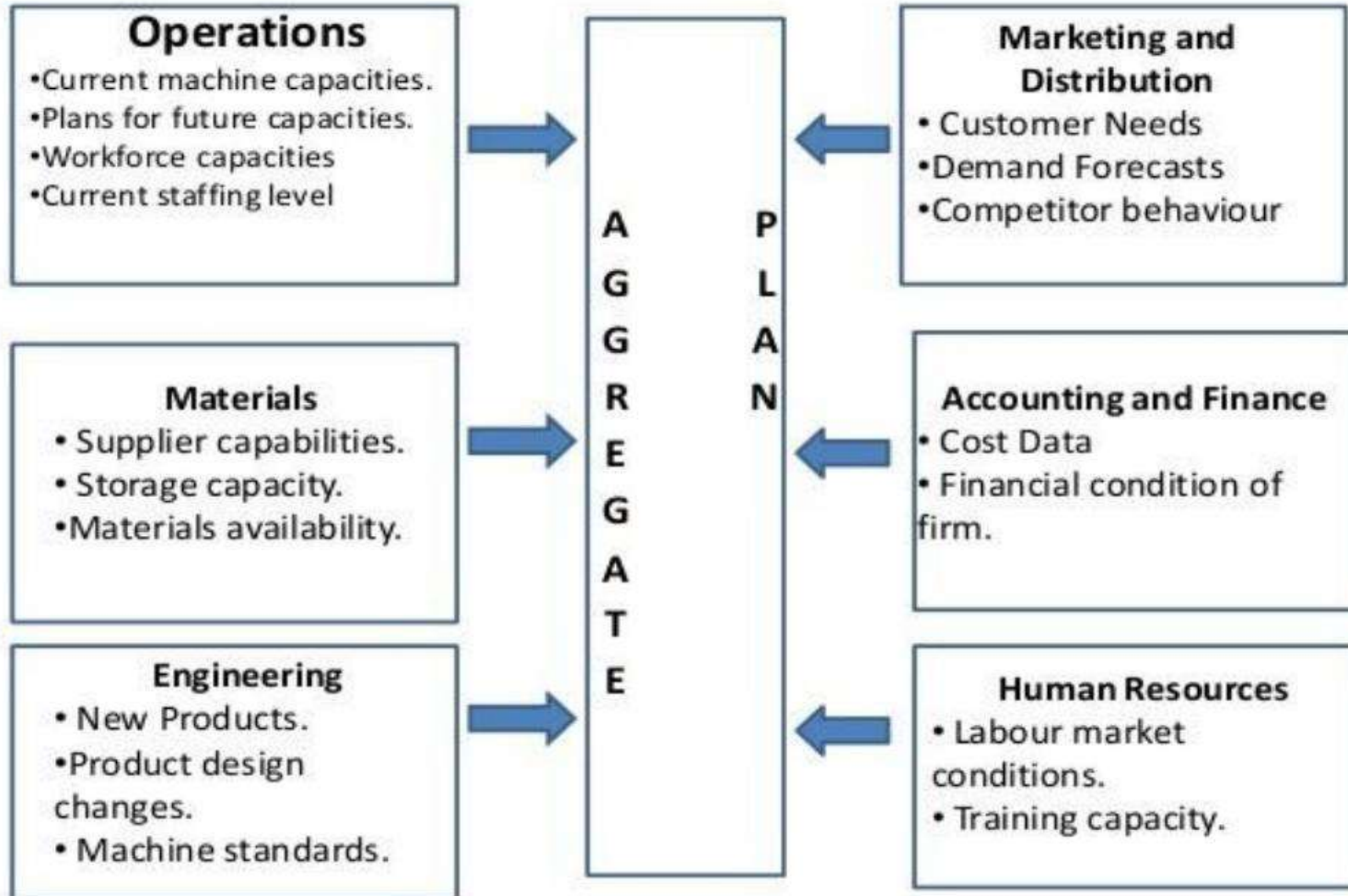
To reduce total cost of operations of the organization to the minimum over that period.

Aggregate planning involves decision making on :

- Quantity of outsourcing,
- Subcontracting of items,
- Overtime of labour,
- Numbers to be hired and fired in each period
- Amount of inventory to be held in stock and to be backlogged for each period.

All of these activities are done within the framework of the company ethics, policies, and long term commitment to the society, community and the country of operation.

AGGREGATE PLANNING PROCESS



Role & Need of Aggregate Planning

- Objective to minimize cost over the planning period by adjusting:
 - Production rates
 - Labor levels
 - Inventory levels
 - Overtime work
 - Subcontracting works
 - Other controllable variables

Advantages of Aggregate Planning

1. Businesses can predict the supply and demand of their products and services.
2. This planning saves money, streamline operations and increase productivity.
3. Helps to develop a game plan that will assist them with determining their staffing requirements, materials needed, estimated timelines and budget costs so they can better plan ahead.
4. Planning for changes in demand months in advance ensures that the change of production schedules can occur with little effort.

Aggregate Planning Strategies

Aggregate Plan Strategies- Level plans

- Use a constant workforce & produce similar quantities each time period
- Use inventories and back-orders to absorb demand peaks & valleys
- Use inventories in better way to absorb the peak of demand and valleys

Aggregate Plan Strategies- Chase plans

- Minimize finished goods inventories by trying to keep pace with demand fluctuations
- Matches demand varying either work force level or output rate.

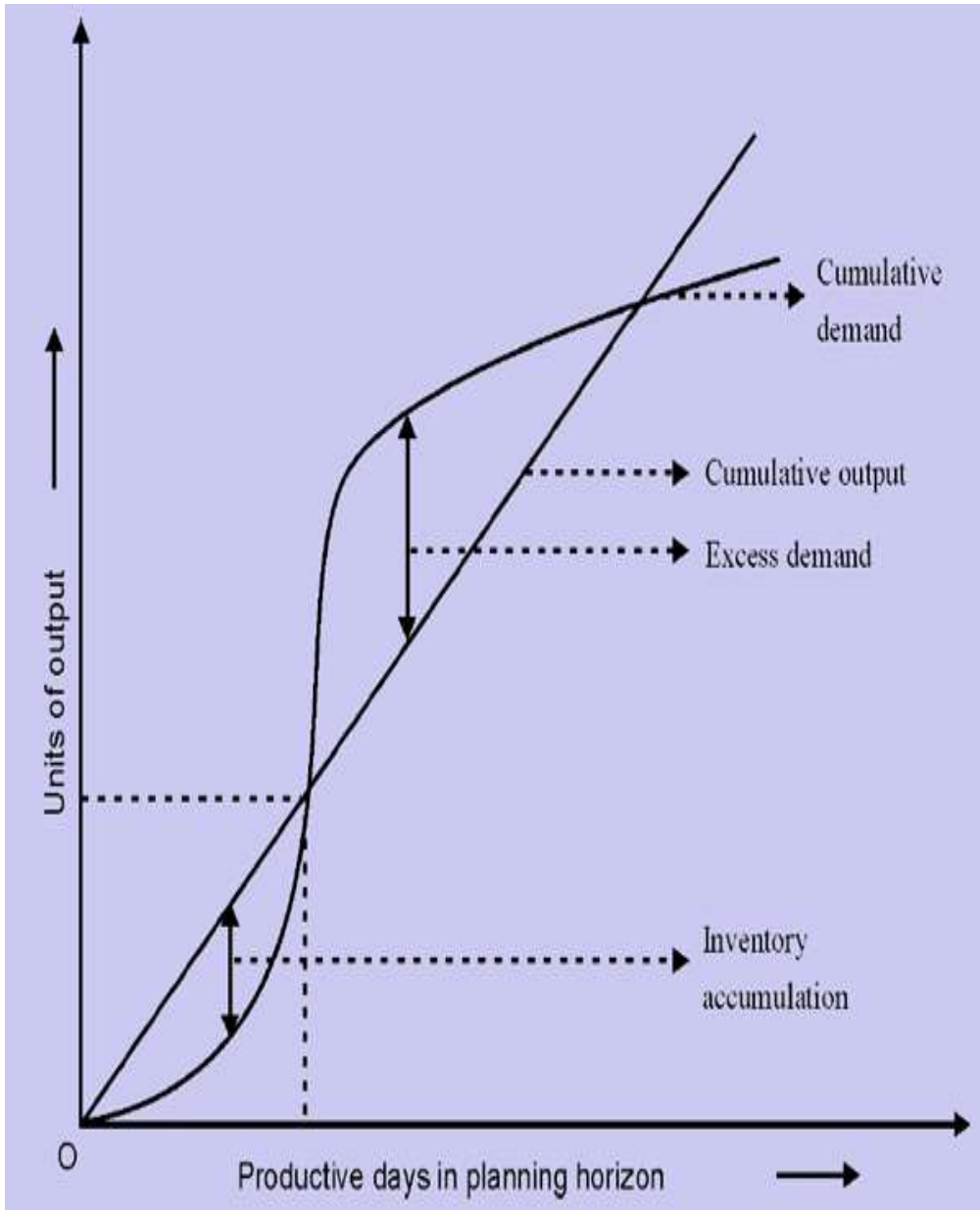
Methods of Aggregate Planning

Graphical Method:

- This method is the Popular techniques.
- Easy to understand and use.

Inputs :

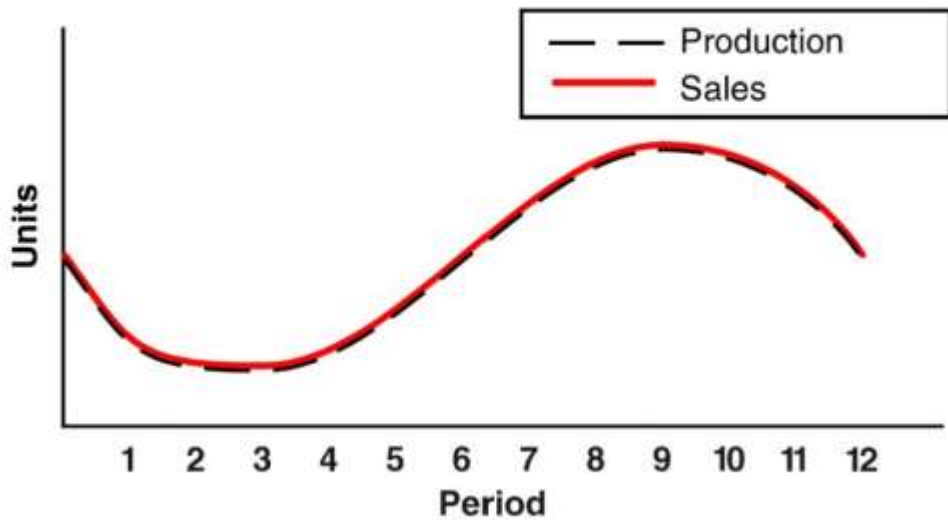
- Determine the demand for each period
- Determine the capacity for regular time, overtime, and subcontracting each period.
- Find labor costs, hiring and layoff costs, and inventory holding costs.
- Consider company policy on workers and stock levels.
- Develop alternative plans and examine their total costs.



Graphical Method for Aggregate Output Planning

- The cumulative productive days for the planning time period on the X axis, & cumulative units of output on the Y axis
- Cumulative demand forecast for the entire planning time period is plotted on the graph.
- Based on the aggregate planning goals, a planning strategy is selected. Proposed output for each period in the planning horizon is computed and plotted on the same axis used to plot the demand.
- Planned output is compared with expected demand

Chase Strategy



Reactive Rate Method

- Keeps inventory levels low until customer demand justifies an increase
- Hires, fires, and lays off employees as necessary based on production levels
- Introduces new strategies and methods of production when demand increases
- Businesses may have problems with fluctuating quality levels due to staffing changes, and they may have conflicts with employees and labor unions due to the frequent need for layoffs.

SCHEDULING - Determine the timing and order of operations, to optimize the use of resources, to meet production requirements

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Module 3

SCHEDULING

- ❑ Scheduling can be defined as “prescribing of when and where each operation necessary to manufacture the product is to be performed”.

OR

- ❑ It is also defined as “determination of time that is required to perform each operation and also the time required to perform the entire series of operations as routed”.

SCHEDULING

- Scheduling is the process of planning, coordinating and controlling the use of resources to complete a production process
- Scheduling helps to streamlining order production based on due date and sequencing of various job work
- Scheduling is used to allocate plant and machinery resources, plan human resources, plan production process and purchase materials

SCHEDULING DECISIONS

- The product that is planned for production
- The configuration of the product
- The quantities that are involved in the production
- The dates when the production will start and end
- The capacity reservations for the resources that perform the production activities

PRINCIPLE OF SCHEDULING

The principle of optimum task size

.scheduling tends to achieve maximum efficiency when the task sizes are small and all tasks of same order of magnitude

Principle of optimum production planning

. The planning should be such that it imposes an equal load on all plants

Principle of optimum sequence

. Scheduling tends to activate maximum efficiency when the work is planned so that work hours are normally used in the same sequence

Priority rules

- First Come First Serve (FCFS)
- Shortest Processing Time (SPT)
- Earliest Due Date (EDD)
- Slack Time Remaining (STR) = time remaining before due date - remaining processing time, The shortest STR goes first.
- Critical Ratio (CR) = Time remaining before due date/Remaining processing time . The smallest CR goes first

Types of scheduling

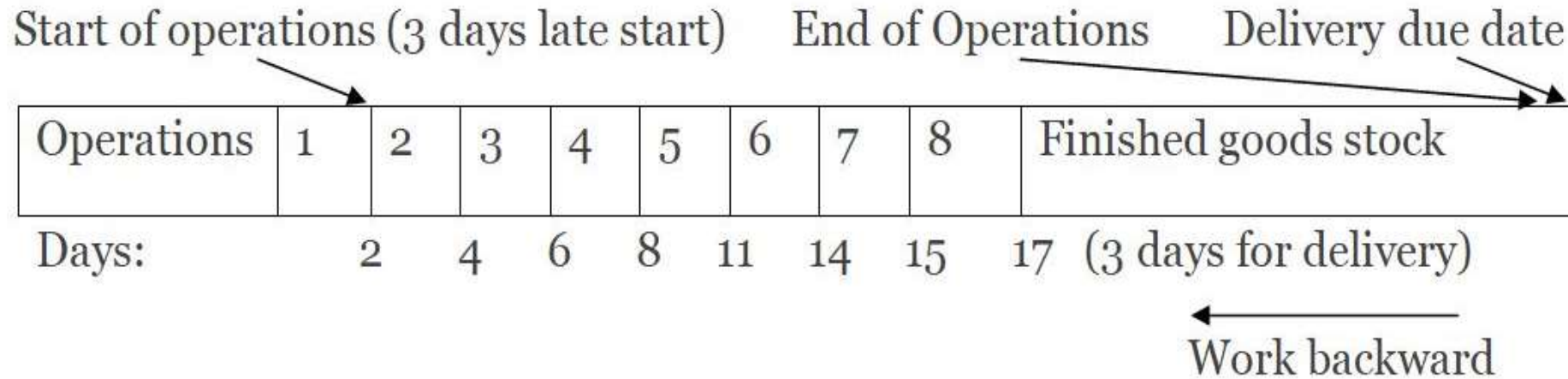
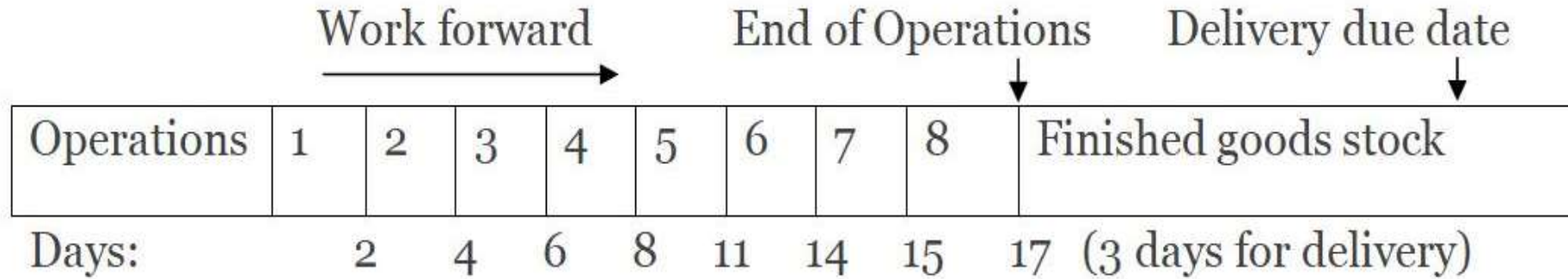
1. Forward scheduling

- It is commonly used in job shops where customers place their order on “needed as soon as possible” basis.
- It determines start and finish times of next priority job by assigning it the earliest available time slot and from that time, determines when the job will be finished in that work centre.

2. Backward scheduling

- It is often used in assembly type industries and commit in advance to specific delivery dates. Backward scheduling determines the start and finish times for waiting jobs by assigning them to the latest available time slot that will enable each job to be completed just when it is due, but done before.

Forward and Backward scheduling



Gantt Chart:

- ❑ A Gantt chart is a project management tool assisting in the planning and scheduling of projects of all sizes, although they are particularly useful for simplifying complex projects.
- ❑ This Gantt chart is used to organize and clarify actual or intended use of resources within a time framework
- ❑ It helpful in scheduling, despatching and control.

Gantt Chart

Task Name	Q1 2019			Q2 2019		Q3 2019
	Jan 19	Feb 19	Mar 19	Apr 19	Jun 19	Jul 19
Planning						
Research						
Design						
Implementation						
Follow up						

Benefits of Gantt Chart

- ❑ Efficiency. It can calculate realistic project completion times and set goals based on available resources.
- ❑ Teamwork. It can access to the same information to all team members.
- ❑ Tracking. Analyze workflows for constraints and adjust their work accordingly.
- ❑ Versatility. They prompt managers to estimate which resources are needed when.
- ❑ Visualization. Provide a project roadmap at a glance.

Index method

- ❑ Used for the purpose of the loading and also for allocating the different jobs to the different machines.
- ❑ Generally orders are assigned on the basis of the “first come first assigned” met
- ❑ Assigning a job to the best machine until its capacity is consumed totally. After this, the rest jobs are transferred to the next available best machine.

Critical Ratio Scheduling

- It is a technique for establishing and maintaining priorities among the jobs in a factory.
- Critical ratio which is an index sets relationship between when a product required and when it can be supplied.

$$\text{Critical ratio} = \frac{\text{Time the job is needed (demand time)}}{\text{Supply lead time}}$$

If $CR > 1$, job ahead of schedule

If $CR < 1$, job behind of schedule

If $CR = 1$, job on schedule

Thank You